

Practical No. 1

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Aim : To write a problem statement and define the project title with a bounded scope for a software project.

1. Selected SIH Problem Statement

Problem Statement ID	26207
Original Title	Student Innovation-Smart education,a concept that describes learning in digital age. It enables learners to learn more effectively, efficiently, flexibly and comfortably.
Organization	AICTE
Department	AICTE, MIC-Student Innovation
Category	Software
Theme	Smart Education

2. Proposed Project Title

Smart Education Platform for Personalized and Flexible Digital Learning

3. Problem Statement

The provided problem statement describes smart education as a concept for learning in the digital age. It focuses on enabling learners to learn more effectively, efficiently, flexibly and comfortably through technology-enabled education.

Traditional learning environments may not provide the same level of flexibility, convenience and personalized access that digital learning can offer. Learners may need educational resources that can be accessed according to their pace, schedule and learning needs.

The proposed Smart Education Platform will provide a software-based learning environment in which students can access digital educational content, organize learning activities, monitor progress and receive a more flexible learning experience. The platform can be designed to support personalized learning paths, convenient resource access and progress tracking.

The system will focus on improving the learning experience through a centralized digital platform while keeping the project scope practical for an academic software project.

4. Problem Being Solved

How can a software-based smart education platform make learning more effective, efficient, flexible and comfortable for learners in the digital age?

The system therefore needs to provide convenient access to learning resources, support flexible study, organize educational activities, track learner progress and present useful information in an understandable digital interface.

5. Objectives of the Project

1. Provide a centralized digital platform for accessing educational content.
2. Enable learners to study at their own pace and according to their schedules.
3. Organize courses, learning resources, assignments and related activities.

4. Support personalized learning through learner-specific progress and recommendations.
5. Track learner activity and progress through a dashboard.
6. Improve convenience and accessibility of digital learning resources.
7. Provide a simple and user-friendly interface for students and educators.
8. Demonstrate how software technology can support smart education.

6. Users / Stakeholders

6.1 Student / Learner

Access courses and digital resources; view lessons; complete learning activities; submit assignments or quizzes; monitor personal progress; and study according to a flexible schedule.

6.2 Teacher / Educator

Create or manage learning content; organize lessons and activities; monitor student progress; and provide educational guidance or feedback.

6.3 Administrator

Manage users, courses, learning resources, access permissions and overall system configuration.

6.4 Educational Institution

Uses the platform to support digital learning, improve accessibility of educational resources and obtain learning-related insights.

7. Core Features of the Smart Education Platform

1. User registration and secure login for students, teachers and administrators.
2. Course and subject management.
3. Digital learning content such as notes, videos, documents and quizzes.
4. Personalized learning dashboard for each learner.
5. Progress tracking based on completed lessons and activities.
6. Assignment and quiz management.
7. Notifications and reminders for important learning activities.
8. Basic analytics and reports for learners and educators.
9. Search and organized access to learning resources.
10. Responsive web-based interface for convenient access across devices.

8. Scope of the Project

8.1 In-Scope

1. Web-based smart education platform prototype.
2. Student, teacher and administrator login and role-based access.
3. Course, subject and learning-resource management.
4. Digital lessons, notes, videos or documents within the platform.
5. Assignments, quizzes and learner activity tracking.
6. Personalized dashboard and progress visualization.
7. Basic notifications and reminders.
8. Basic reports and analytics related to learning progress.
9. Database storage for users, courses, resources and learner activity.
10. Testing using sample or simulated educational data.

8.2 Out-of-Scope

1. Nationwide or institution-wide production deployment.
2. Replacement of complete university or school management systems.
3. Development of advanced artificial general intelligence for teaching.
4. Full-scale live examination proctoring using specialized hardware.
5. Integration with every external learning management or payment platform.
6. Guaranteed personalization accuracy for every learner.
7. Complete offline functionality for all platform services.
8. Development of a commercial-scale enterprise education ecosystem.

9. Functional Requirements

ID	Functional Requirement
FR01	The system shall allow users to register and log in according to their assigned role.
FR02	The system shall provide role-based access for students, teachers and administrators.
FR03	The system shall allow teachers or administrators to create and manage courses and learning resources.
FR04	The system shall allow students to browse and access available learning content.
FR05	The system shall provide assignments, quizzes or other learning activities.
FR06	The system shall record learner activity and completion status.
FR07	The system shall display learner progress through a dashboard or report.
FR08	The system shall support basic personalized learning information or recommendations.
FR09	The system shall provide notifications or reminders for relevant learning activities.
FR10	The system shall store user, course, resource and activity information in a database.
FR11	The system shall allow authorized users to update or remove managed content.
FR12	The system shall provide basic reports or analytics for learning progress.

10. Non-Functional Requirements

Usability: The interface should be simple, clear and easy for learners and educators to understand.

Performance: The system should load common pages and learning resources within acceptable response times for the defined academic prototype.

Security: User authentication and role-based authorization should protect learning data and administrative functions.

Availability: The platform should remain available during normal learning and demonstration periods.

Scalability: The design should allow the number of users, courses and resources to increase within defined system limits.

Reliability: The system should save learning activity and progress consistently and avoid data loss during normal operation.

Maintainability: The software should be modular enough to allow future addition of courses, learning features and personalization logic.

Accessibility: Learning resources and interface elements should be presented in a way that supports convenient digital access across common devices.

11. Basic System Boundary

User → Authentication → Role-Based Dashboard → Course / Content Management → Learning Activities → Progress Tracking → Personalized Learning Information → Reports / Notifications

The project boundary covers the smart education software platform and its associated learning-management functions. External enterprise-scale education systems and complete commercial ecosystems are outside the prototype boundary.

12. Assumptions

1. Users have access to a suitable internet-connected device for the web-based prototype.
2. Educational content is available in digital form for demonstration.
3. Students and educators provide valid information when using the system.
4. The prototype is developed and tested using sample educational data.
5. The institution or project team provides the required computing and hosting resources.
6. The effectiveness of personalized learning depends on the quality of available learner and course data.

13. Limitations

1. The prototype may not cover every type of educational activity used by real institutions.
2. Personalization quality may be limited when only a small amount of learner data is available.
3. System performance may differ between a laboratory prototype and a large production deployment.
4. Dependence on internet connectivity may affect access to online resources.
5. The project does not guarantee improved academic performance for every learner.
6. Advanced AI tutoring, large-scale proctoring and complex third-party integrations are outside the bounded scope.

14. Bounded Scope Summary

Area	Included	Excluded
Learning	Digital and flexible learning support	Replacement of all classroom teaching
Users	Students, teachers and administrators	Nationwide public education identity systems
Content	Courses, notes, videos, documents, quizzes	Every possible external content platform
Personalization	Basic learner-specific progress and recommendations	Guaranteed advanced AI tutor for every learner
Assessment	Assignments, quizzes and progress records	Full-scale high-stakes examination infrastructure
Interface	Responsive web dashboard	Complete commercial LMS ecosystem
Environment	Academic prototype / controlled testing	Nationwide production deployment
Data	Application, course and learner activity data	Unlimited real-world educational data integration

15. Conclusion

The problem statement and bounded scope of the Smart Education Platform for Personalized and Flexible Digital Learning were defined from the provided SIH problem statement 26207. The project addresses the need for learning in the digital age by focusing on effectiveness, efficiency, flexibility and comfort for learners.

The proposed software platform brings learning resources, activities, progress tracking and basic personalization into a single digital environment. Its scope is intentionally bounded to make the project achievable as an academic software engineering practical while still demonstrating the main idea of smart education.

This practical provides a foundation for subsequent Software Engineering activities such as requirement engineering, SRS preparation, use-case modelling, activity diagrams, DFDs, UML diagrams, database design, testing, risk analysis, scheduling and software quality assurance.